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June 15, 1967
INTERIM REVISION OF
Fed. Spec. W-L-310a
November 14, 1955

INTERIM FEDERAL SPECIFICATION

LIGHT, DESK (FLUORESCENT PEDESTAL)

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification W-L-310a, dated November 14, 1955.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers freestanding fluorescent pedestal desk lights, without fluorescent lamps. Desk lights for use in hazardous locations are not included.

1.2 Classification. The fluorescent pedestal desk light furnished under this specification shall be the type that will accomodate two, 15-watt, 18-inch, T-8 fluorescent lamps that conform to W-L-116.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:
- J-C-580 - Cord, Flexible, and Wire, Fixture, (Electrical, 0-to 600-Volt Service).
- W-B-30 - Ballast, Fluorescent Lamp.
- W-L-116 - Lamps, Fluorescent.
- W-S-610 - Splice, Conductor.
- PPP-B-585 - Boxes, Wood, Wirebound.
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Box, Fiberboard.
- PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall.

Federal Standard:
- Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
- MIL-P-116 - Preservation, Methods of.

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. Standards (UL):
UL-153 - Portable Electric Lamps.
UL-542 - Lampholders, Starters, and Starter Holders for Fluorescent Lamps.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, New York 11749; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050).

3. REQUIREMENTS

3.1 Each contractor shall submit to the contracting agency proof that the desk light he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc., UL-153, Standard for Portable Electric Lamps. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc. may be accepted as evidence that the desk light conforms to this requirement.

3.1.1 In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the desk light conforms to the applicable requirements of the published standards including methods of test of the Underwriters' Laboratories, Inc., UL-153.

3.1.2 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.2 Design.

3.2.1 Current. Unless otherwise specified (see 6.1), the desk lights shall be designed for operation on 110-125-volt, 60-cycle, alternating-current.

3.2.2 Material and devices.

3.2.2.1 Material. The material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market. The material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

3.2.2.2 Devices. The desk lights shall be furnished complete with reflector, ballast, lampholders, switch, wiring and all other devices necessary for operation except fluorescent lamps.

3.2.3 Construction. The desk lights shall be constructed of metal. Bases, columns, and supports shall be fabricated from one or more of the following: iron, steel, brass, bronze, aluminum, aluminum alloy, zinc alloy, or magnesium alloy. White-metal castings (lead base) shall not be used in bases, columns, or supports. Each desk light shall consist of a metal base with one or more metal columns or other supports, on the upper end of which shall be mounted a reflector and reflector housing. The reflector may be integral with the housing or may be separate and enclosed by a housing.

3.2.4 Exposed edges. Exposed edges of sheet metal or other metal parts shall be so finished that personal injury due to sharp edges will not occur under any normal circumstances.

3.2.5 Wiring. The only exposed wiring shall be from the base to the plug connector. All other wiring shall be entirely concealed.

3.2.6 Base. The base shall be of cast, pressed, or spun metal and the minimum wall thickness shall be as follows:

Gravity cast metal base	- 0.156 inch
Pressure cast metal base	- 0.065 inch
Pressed or spun metal base	- 0.0359 inch

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The lamp ballast shall be enclosed within the base, and the base area and weight shall be sufficient to cause the light to be freestanding on a horizontal surface. When inclined at an angle of 8 degrees from the horizontal in any direction, the light shall not overturn. The bottom of the base shall be a plane surface and shall be covered with a dark-colored felt at least 1/16 inch thick or provided with other suitable means to prevent scratching of the surface on which the light rests. If an adhesive is used to attach the felt or other suitable material to the base, there shall be no evidence of penetration or wicking.

3.2.7 Column(s). The desk light shall have one or more columns of cast metal, formed sheet metal, or flexible metal tubing. The columns may be attached at the base and the reflector housing through jointed connections (see 6.1). The cross-sectional area of the column(s) shall be sufficient to house all internal wiring. The wall thickness and the method of attachment to the base and the reflector housing shall be sufficient to provide the strength and rigidity required to successfully fulfill the test requirements of this specification.

3.2.8 Reflector housing and reflector. The reflector housing and the reflector shall consist of an integral metal piece or separate metal pieces containing the lampholders. The reflecting surface shall be provided on a separate metal reflector or directly on the inside surface of the housing. Separate metal reflectors shall be of steel or aluminum with a reflecting surface to provide required illumination levels in 3.3.1, and shall be held rigidly within the housing. The housing shall be of steel or aluminum and shall be of rigid construction. The reflector housing shall be joined to the supporting column(s) rigidly or by means of a jointed connection.

3.2.9 Ballast. The ballast furnished with the desk light shall be suitable for operation on the current specified and shall be in accordance with W-B-30.

3.2.10 Lampholders. Each lampholder shall be so constructed and mounted as to permit easy insertion and removal of a 15-watt, 18-inch, T-8 fluorescent lamp. The lampholders shall be in accordance with the applicable requirements of UL-542.

3.2.11 Switch. Each desk light shall be provided with an operating switch mounted in the base or in the reflector housing. The switch shall be in accordance with the applicable requirements of UL-153.

3.2.12 Plug connector and cord. A cord having an exposed length of not less than 7 feet with a molded-on, parallel blade plug connector shall be furnished with the desk light. The cord shall be in accordance with type SP-2 or SPT-2 of J-C-580.

3.2.13 Splices. All splice connections shall be made with splice conductors in accordance with class 1, style G of W-S-610.

3.3 Performance.

3.3.1 Illumination. The light shall provide the following minimum illumination on a horizontal surface when tested as specified in 4.4.1.

Distance from the center of lamps Inches	Illumination on horizontal surface Foot-candles
12	70
18	35
24	17
30	8
36	4

3.3.2 Ratio of variation in illumination. When the light is on a horizontal surface, with the reflector in the position specified in 4.4.1, the maximum permissible ratio of the illumination between a point on the surface 30 inches distant therefrom on a line perpendicular to a vertical plane through the longitudinal axis of the lamps, shall be 18 to 1.

3.3.3 Angle of cutoff. The light shall be so constructed that, with the reflector in the position specified in 4.4.1, no part of the lamps or reflecting surface shall be visible when viewed from a point one foot above the horizontal surface and two feet from a vertical plane through the longitudinal axis of the lamps.

3.3.4 Rigidity. The light shall be so constructed that no permanent deformation shall result when the light is subjected to the test specified in 4.4.3.

3.4 Finish. All exposed metal parts shall be cleaned and free of dirt, oil, grease, and moisture before applying the finish. All surfaces except aluminum reflecting surfaces, shall be covered completely with one or more coats of enamel or by electro-plating. Reflecting surfaces other than aluminum shall be white and all other enameled surfaces shall be either smooth or crackle finish

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of brown, bronze, or gray, as specified in the invitation for bids (see 6.1). All enameled surfaces shall be free from tears, star marks, crazing, blisters, hairlines, chipping, pinholes, and other irregularities or defects.

3.5 Nameplate. Unless otherwise specified in the invitation for bids (see 6.1), a nameplate permanently and legibly providing the following information shall be securely attached to the desk light in an inconspicuous location:

Specification title and number
Current, voltage, and frequency
Manufacturer's model number
Manufacturer's name or trade-mark.

3.6 Workmanship. The desk lights shall be free from defects which detract from the appearance or which might impair the serviceability. The entire reflector and light assemble shall be firmly seated at all points and locked with nuts and washers wherever necessary.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection. Sampling for visual and dimensional inspection shall be conducted in accordance with MIL-STD-105, inspection level II. Acceptance shall be based on an acceptable quality level (AQL) of 2.5 defects per hundred units.

4.2.2 Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-2. Acceptance shall be based on an AQL of 1.5 defects per hundred units.

4.3 Inspection. The desk light shall be inspected for compliance with the following requirements:

Current (see 3.2.1)
Exposed edges (see 3.2.4)
Wiring (see 3.2.5)
Base (see 3.2.6)
Joints and connections (see 3.2.7 and 3.2.8)
Lampholders (see 3.2.10)
Switch (see 3.2.11)
Plug connector and cord (see 3.2.12)
Finish (see 3.4)
Name plate (see 3.5)
Workmanship (see 3.6).

4.4 Tests.

4.4.1 Photometric. The sample desk light shall be equipped for test with the required number of 15-watt, 18-inch, daylight-type, T-8 fluorescent lamps which have been aged not less than 100 hours and which have not had more than 500 hours of use. The sample desk light shall be placed on a horizontal surface representing a desk or table top with the front edge of the reflector horizontal. Samples of desk lights with moveable reflector housings shall be positioned so that the front and back edges of the reflector housing are in a plane parallel to the horizontal surface on which the desk light rests and at a vertical distance of 12 inches fron the horizontal surface or at the greatest possible vertical distance less than 12 inches which satisfies these conditions. A Macbeth illuminometer or other suitable instrument capable of being read in foot-candles within the range of illumination values given in 3.3.1 and 3.3.2 and which has been calibrated to indicate the values of illumination of fluorescent light shall be used. The readings shall be taken perpendicular to the normal of the lamp and, if a cosine corrected meter is not used, shall be multiplied by the cosine of the angle between the vertical and the direction of the normal to the lamps to obtain the illumination on the horizontal surface. It is often more convenient to measure the angle between the normal to the lamps and the horizontal; if this angle is used, the readings shall be multiplied by the sine of the angle in order to obtain the horizontal reading.

4.4.2 Dielectric strength. Lampholders shall withstand a dielectric strength test consisting of the application for one minute of an alternating current having a sine wave shape and an effective

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voltage of 1,000 volts plus twice the rated voltage of the lamp. The test shall be made between terminals of opposite polarity, between current-carrying parts and grounded metal, or between current-carrying parts and metal parts exposed on the surface of the mounting.

4.4.3 Rigidity.

4.4.3.1 Fixed column(s). The sample light shall be placed on its base on a horizontal surface with the reflector in its normal position. The vertical distance from the horizontal surface to the two points of the reflector housing farthest from the center of the support of the column(s) shall be measured. A 5-pound weight shall be suspended for one minute, first from one of the points and then from the other. The sag of the reflector housing with the weight suspended shall not exceed 1/4 inch. After removal of the weight, the vertical distance of the two points shall not disclose any permanent deformation or bending of the reflector housing or column(s). Consideration shall be given to the compressibility of the base covering during the conducting of the test. The base may be clamped as necessary to prevent tilting.

4.4.3.2 Moveable and flexible column(s). With the reflector housing and column(s) in the same position used in conducting the photometric test in 4.4.1, the light shall be placed on a horizontal platform and the base securely clamped to the surface. A load equal to three times the combined weights of the reflector, reflector housing, and lamps shall be suspended from the reflector housing at its midpoint for a period of one minute. Movement of the column(s) shall not cause the reflector housing to sag more than 1/4 inch while the weight is suspended.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Table I - Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Level A. Each light shall be packaged in accordance with method IA-15 of MIL-P-116. Die-cut blocking and bracing shall be provided.

5.1.2 Level B. Each light shall be packaged in a close-fitting box conforming to PPP-B-636, class-domestic. Die-cut blocking and bracing shall be provided. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. The lights shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Unit packages of lights of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, type III; PPP-B-621, class 2; PPP-B-636, class-weather resistant; or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Six unit packages of lights of like description shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

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5.3 Standard Pack (Civil Agency Procurements). The Standard Pack for civil agency procurements shall be as specified in 5.1.2 and 5.2.2 (see 6.2).

5.4 Marking.

5.4.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Current rating, if other than specified (see 3.2.1).
(c) Finish required (see 3.4).
(d) Nameplate information, if other than specified (see 3.5).
(e) Column(s) requirements (see 3.2.7).
(f) Responsibility for inspection, if other than specified (see 4.1).
(g) Level of packaging and packing required (see 5.1 and 5.2).
(h) Special marking, if required (see 5.4).

6.2 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurement of Stores Stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

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